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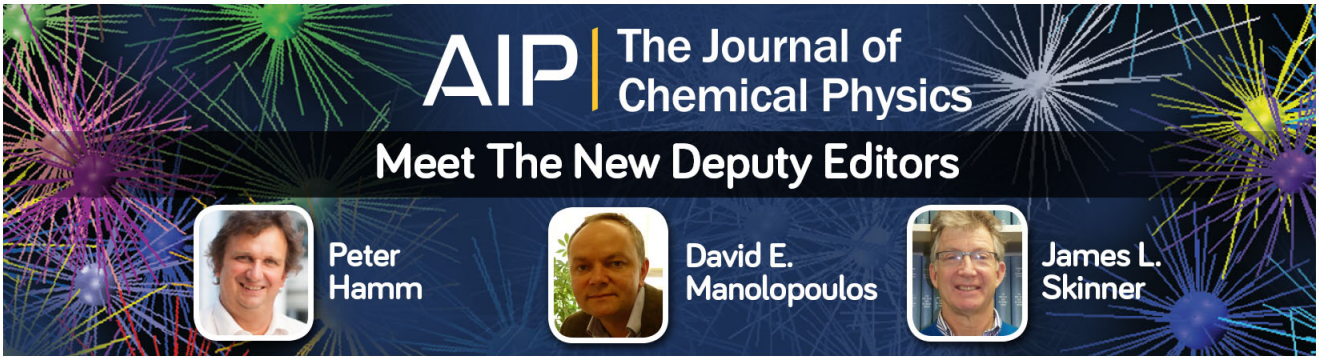

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Electronic excitations of C_{60} fullerene calculated using the *ab initio* cluster expansion method

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The electronic excited states and optical absorption spectrum of C_{60} fullerene below 6.2 eV (200 nm) were calculated using the *ab initio* many-body wavefunction theory of cluster expansion method: the symmetry-adapted cluster-configuration interaction method. Not only optically allowed states but also optically forbidden states were calculated for studying the observed weak absorptions in the visible region. The lowest calculated singlet excited state was the 1^1G_g state. The electron correlation effects are important in determining the energy levels of the four low-lying states that have the character of degenerated HOMO–LUMO transition. The lowest optically allowed 1^1T_{1u} state was calculated at 3.67 eV; this is significantly higher than the energy values found in previous density functional calculations. The observed weak absorption around 3.08 eV appears to correspond to the optically forbidden 1^1T_{2u} state with intensity borrowing via vibronic couplings. © *2012 American Institute of Physics*. [http://dx.doi.org/10.1063/1.4757066]

I. INTRODUCTION

Because of their scientific and practical importance, fullerenes are major target molecules for quantum chemical calculations.[1] Their unique properties are largely attributable to their electronic structure, including their excited states. However, high-accuracy *ab initio* calculations have not been reported for the electronic excited states of C_{60}. The assignments of ultraviolet–visible (UV–vis) spectra of fullerenes have been provided based on semiempirical methods or time-dependent density functional theory (TDDFT) calculations.[2] Such methods may be useful, but parameter or functional dependences cannot be ruled out. Therefore, the need for high-level *ab initio* calculations of many-body wavefunction theories is quite compelling.[3] The molecular size and their conjugated electronic structure make practical computations for fullerenes difficult. Efficient schemes that rely on the locality of electronic structures may not be adequate for such systems. Furthermore, many excited states must be calculated to reproduce the UV–vis spectrum, because there are many low-lying electronic states. The latter problem can be avoided if the symmetry reduction of the I_h point group for C_{60} is used fully. However, such computational programs are not widely available. High symmetry cannot be used for extended applications such as C_{70} fullerene, substituted fullerenes, photochemical additions to fullerenes,[4] and fullerenes encapsulating a molecule.[5]

Recently, we found that the direct symmetry-adapted cluster-configuration interaction (SAC-CI) method[6] is useful for the calculations of electronic excited states of large conjugated systems.[7] The code of the direct SAC-CI method has been released in GAUSSIAN 09 B01 and later.[8] In this study, we used its development version in which the computational efficiency of SAC-CI was improved, mainly by using parallel computing schemes. Recently, the static and dynamic polarizabilities of C_{60} in the ground state were calculated by the linear response (LR) coupled cluster with singles and doubles (CCSD), which is closely related to the SAC-CI, using parallel computation.[9] In this study, we report the *ab initio* calculation including the electron correlation effects for the electronic excited states of C_{60} fullerene below 6.2 eV (200 nm) including optically forbidden states. Such optically forbidden states are important not only for the applicability and reactivity of C_{60} but also for the absorption spectrum of C_{60} because there are many vibronically induced bands with false origins. Quantum chemical computations are useful for studying optically forbidden states. We calculated the low-lying 61 excited states of C_{60} in I_h symmetry, which corresponds to 230 states in the C_{2h} symmetry used in the actual calculation, in the UV–vis region. The optical absorption spectrum of C_{60} was assigned based on the SAC-CI results.

The low-lying HOMO-LUMO transitions of C_{60} observed in visible region are optically forbidden. Therefore, the previously proposed assignments of electronic spectra were given based on the observed vibronic bands and the computed intensities of forbidden states. In such studies, the intensity borrowings were calculated based on the Herzberg–Teller couplings using semiempirical wavefunctions of the complete neglect differential overlap for spectroscopy (CNDO/S).[10] By the analysis of the vibronic structure of the fluorescence and fluorescence-excitation spectra, van den Heuvel *et al.* proposed that the emitting lowest singlet excited state has T_{1g} character dominantly, but G_g character is appreciably mixed.[11] Negri *et al.* proposed that the lowest state is 1^1T_{1g} and the second lowest state is 1^1G_g that is about 120 cm^{-1} (0.015 eV) above the 1^1T_{1g} state.[12] Sassara *et al.* simulated

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the fluorescence spectrum using the calculated vibronic transition intensities and the observed spectrum was well reproduced by mixing the 1^1T_{1g}, 1^1T_{2g}, and 1^1G_g states.[13] These studies showed that experimental spectra can be explained based on the vibronic intensities calculated with CNDO/S wavefunction and the assumption that the 1^1T_{1g} is the lowest state. These findings are consistent with a magnetic circular dichroism (MCD) spectrum.[14] However, the calculated energy levels of the excited states by CNDO/S were 1^1T_{2g} (2.29 eV) < 1^1T_{1g} (2.33 eV) < 1^1G_g (2.34 eV) < 1^1H_g (2.65 eV); the calculated CNDO/S energies were inconsistent with the analyses of vibronic bands.[10] Therefore, the accuracy of the CNDO/S wavefunctions seems to be problematic. To our knowledge, few *ab initio* calculations with electron correlations or the TDDFT calculations have been reported about the optically forbidden excitations of C_{60} in the visible region.[15] Therefore, the reliability of using the semiempirically obtained results has not been discussed in terms of the comparison with *ab initio* results. By the present SAC-CI calculation, the lowest singlet excited state was the 1^1G_g state. We found the importance of electron correlation for the lowest HOMO–LUMO transitions.

A weak band (band A) is observed around 400 nm. Based on the resonance-enhanced two-photon ionization (R2PI) spectra, Haufler *et al.* suggested that this band is a vibronically induced transition of an optically forbidden state.[16] However, semiempirical[17] and TDDFT[18] calculations assigned this band to the optically allowed 1^1T_{1u} state, although the calculated intensity appeared to be overestimated. By the present SAC-CI calculation, the lowest optically allowed 1^1T_{1u} state was calculated at 3.67 eV; this is significantly higher than the energy value found in previous density functional calculations. We assigned the optically forbidden 1^1T_{2u} state to the band A observed around 3.08 eV. Consequently, the present SAC-CI results could consistently explain the experimental findings. The SAC-CI results were compared with the TDDFT calculation with PBE0 functional. Based on the SAC-CI results, we found that the hybrid TDDFT results are more reliable than pure DFT results.

II. COMPUTATIONAL DETAILS

The ground and singlet excited states of C_{60} were calculated using the SAC and SAC-CI methods.[19,20] The wavefunction of SAC-CI for the pth electric state may be written as

$$\Psi^{(p)}_{\text{SAC-CI}} = \sum_K d_K^{(p)} R_K^\dagger \Psi_{\text{SAC}}, \tag{1}$$

where $R_K^\dagger$ represents a symmetry-adapted singlet excitation operator with coefficient d_K. The SAC wavefunction for the ground state is described by the cluster-expansion formalism:

$$\Psi_{\text{SAC}} = \exp\left(\sum_I c_I S_I^\dagger\right)\Phi_0, \tag{2}$$

for the closed-shell Hartree–Fock Φ_0. Singles (S^1) and doubles (S^2) are included for the cluster operators $\{S_I^\dagger\}$:

$$S^1 = \sum_j^{\text{occ}} \sum_b^{\text{vac}} (a_{b\alpha}^\dagger a_{j\alpha} + a_{b\beta}^\dagger a_{j\beta})/\sqrt{2} \tag{3}$$

and

$$S^2 = \sum_{jk}^{\text{occ}} \sum_{bc}^{\text{vac}} (a_{b\alpha}^\dagger a_{j\alpha} + a_{b\beta}^\dagger a_{j\beta})(a_{c\alpha}^\dagger a_{k\alpha} + a_{c\beta}^\dagger a_{k\beta})/\sqrt{2}, \tag{4}$$

where $a_{p\omega}$ and $a_{p\omega}^\dagger$ are electron annihilation and creation operators, respectively, for orbital p; $\omega = \alpha$ or β denotes electron spin. For the excited states, singles (R^1) and doubles (R^2) are included as the excitation operators $\{R_K^\dagger\}$:

$$R^1 = \sum_j^{\text{occ}} \sum_b^{\text{vac}} (a_{b\alpha}^\dagger a_{j\alpha} + a_{b\beta}^\dagger a_{j\beta})/\sqrt{2} \tag{5}$$

and

$$R^2 = \sum_{jk}^{\text{occ}} \sum_{bc}^{\text{vac}} (a_{b\alpha}^\dagger a_{j\alpha} + a_{b\beta}^\dagger a_{j\beta})(a_{c\alpha}^\dagger a_{k\alpha} + a_{c\beta}^\dagger a_{k\beta})/\sqrt{2}. \tag{6}$$

The cluster amplitudes c_I are determined by solving the SAC equations:

$$\langle \Phi_0 | S_K (H - \Delta E_{\text{SAC}}) | \Psi_{\text{SAC}} \rangle = 0, \tag{7}$$

and the excitation coefficients d_K are determined by solving the SAC-CI secular equations:

$$H \left| \Psi^{(p)}_{\text{SAC-CI}} \right\rangle = \Delta E^{(p)}_{\text{SAC-CI}} \left| \Psi^{(p)}_{\text{SAC-CI}} \right\rangle. \tag{8}$$

The energy and wavefunction for many electric states (for numbers of p) can be obtained by this single eigenvalue problem. The dimensions of the SAC and SAC-CI equations are huge for large systems; therefore, iterative methods are employed to obtain the solutions.[6,21]

The equilibrium molecular geometry in the ground state was optimized using the B3LYP/6-31G* method[22,23] in I_h point group symmetry. The calculated C–C distances were 145.3 pm and 139.6 pm for the long and short C–C bonds. They are intermediate values between the electron diffraction[24] (145.8 pm and 140.1 pm) and the x-ray diffraction[25] (145.0 pm and 138.7 pm) experiments. The vertical electronic excitation energies and oscillator strengths were calculated by the SAC/SAC-CI method with D95(d) basis plus diffuse s- and p-type basis[26] placed on the center of five-membered rings with exponent $\alpha(s) = 2.3 \times 10^{-2}$ and $\alpha(p) = 2.1 \times 10^{-2}$ to describe low-lying Rydberg states.

For the SAC/SAC-CI calculation, 120 occupied and 559 unoccupied molecular orbitals (MOs) were used from 180 occupied and 768 unoccupied MOs obtained by the Hartree–Fock calculations; 60 occupied MOs correspond to 1s and 209 unoccupied MOs with orbital energies higher than 2.3 E_h were excluded from the SAC/SAC-CI calculation to reduce the computational requirements. In the SAC/SAC-CI calculation, C_{2h} point group symmetry that is the largest abelian subgroup of I_h was used.

We used the perturbation selection with energy thresholds $\lambda_g = 5 \times 10^{-7}$ and $\lambda_e = 1 \times 10^{-7}$ for the ground

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TABLE I. The dimensions of the SAC/SAC-CI calculation of C_{60}.

	X^1A_g (SAC)[a]	^1A_g	^1B_g	^1A_u	^1B_u
Before selection	562 626 702	562 525 702	562 472 978	562 472 980	562 492 160
After selection	1 438 692	5 716 256	5 917 671	5 273 962	5 368 507

[a]The SAC ground state.

and excited states, respectively. The threshold for the ground state was improved from the standard "LevelThree" sets of GAUSSIAN 09 keyword ($\lambda_g = 1 \times 10^{-6}$ and $\lambda_e = 1 \times 10^{-7}$) because the number of reference states using for the selection of the excited states is much larger than the number of references for the ground state. The excitation operators, S^2 and R^2, having second-order contributions to the energy that were estimated to be smaller than those thresholds were excluded in the practical calculations.[27] The importance of the contributions beyond the second-order was reported for the polarizabilities of C_{60} based on the LR CCSD calculations.[9] From our previous calculations,[6,7] however, this second order energy-based selection technique works well for excitation energies of large conjugated molecules probably because most of the errors in the ground and excited states are cancelled out.

For each symmetry representation, the 80 solutions of the configuration interaction with single excitations (CIS) from the lowest energy were used for the references of the selection scheme and the initial vector for solving the SAC-CI equations. The numbers of excitation operators before and after the perturbation selection are shown in Table I. We did not consider the degeneracy of states for the perturbation selection; therefore, the SAC-CI solutions obtained that should be degenerated in I_h symmetry may have slightly different energies because of the error involved in the selection procedure. Thus, we averaged the energies of such degenerated states and summed the oscillator strengths. We also performed the TDDFT calculations of PBE0 functional[28] with the same basis set and geometry as those of the SAC-CI. Computations were performed with the GAUSSIAN 09 development version[8] on a computer with Intel(R) Xeon(R) X5690 CPUs (6 cores × 2), 198 GB memory, and a 10 TB hard disk.

III. RESULTS AND DISCUSSIONS

The SAC-CI results are summarized in Table II. In I_h symmetry, only the $^1T_{1u}$ states are optically allowed.[31] The

TABLE II. The excitation energies (EE), oscillator strengths, optical absorption wavelengths (abs), and spatial extents of electronic wavefunction ($\langle r^2 \rangle$) of C_{60} calculated by the SAC-CI method.

	SAC-CI				Experiment	TDBP86[a]
State	EE (eV)[b]	Abs. (nm)	$\langle r^2 \rangle$ (Å^2)[c]	Main transitions	EE (band)	EE (state)
1^1G_g	1.75	709	4753.83	H[d] → L[e]	1.82[f] (β_0), 1.91[f] (β_2)	
1^1T_{2g}	1.90	653	4753.86	H → L	2.00[g] (γ_0), 2.08[g] (γ_2)	
1^1T_{1g}	1.94	639	4753.87	H → L	2.10[g] (γ_3), 2,18[g] (γ_5)	
1^1H_g	2.09	593	4753.94	H → L	2.29[g] (δ_1)	
1^1T_{2u}	2.98	416	4753.82	H → L + 1	3.07[g] (A_0)	2.82 (1^1T_{1u})
1^1H_u	3.30	376	4753.95	H → L + 1		
1^1G_u	3.38	367	4754.00	H − 1 → L, H − 2 → L		
2^1G_u	3.39	366	4754.00	H → L + 1		
2^1H_u	3.50	355	4754.04	H − 1 → L		
1^1T_{1u}	3.67 (0.200)	338	4754.12	H → L + 1	∼3.5[f,g] (B)	
3^1H_u	3.68	337	4754.15	H − 2 → L		
3^1G_u	3.69	336	4754.12	H − 2 → L, H → L + 1		
2^1T_{2u}	3.74	331	4754.12	H − 1 → L + 1		
3^1T_{2u}	3.83	324	4754.19	H − 2 → L + 1		
2^1T_{1u}	3.90 (1.265)	318	4754.11	H − 1 → L + 1	3.78,[g] 3.80[h] (C)	3.51 (2^1T_{1u})
4^1H_u	4.15	299	4757.36	H → L + 3		
2^1H_g	4.23	293	4754.42	H → L + 6		
2^1T_{1g}	4.32	287	4754.55	H → L + 6		
2^1T_{2g}	4.43	280	4754.63	H → L + 6		
2^1G_g	4.50	275	4754.43	H → L + 6		
4^1G_u	4.53	274	4754.24	H → L + 7		
5^1H_u	4.58	271	4754.28	H → L + 7		
1^1A_u	4.62	268	4754.32	H → L + 7		
6^1H_u	4.71	263	4754.42	H → L + 7		
3^1T_{2g}	4.74	261	4754.20	H − 2 → L + 1, H − 1 → L + 1		
4^1T_{2u}	4.75	261	4754.34	H → L + 7		
3^1G_g	4.76	260	4754.28	H − 1 → L + 1		

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TABLE II. (*Continued.*)

	SAC-CI				Experiment	TDBP86[a]
State	EE (eV)[b]	Abs. (nm)	$\langle r^2 \rangle$ (Å^2)[c]	Main transitions	EE (band)	EE (state)
5^1G_u	4.76	260	4754.37	H → L + 7		
3^1T_{1g}	4.89	254	4754.30	H − 1 → L + 1		
3^1H_g	4.90	253	4754.26	H − 2 → L + 1, H − 1 → L + 1		
4^1T_{1g}	4.95	250	4754.54	H − 2 → L + 1		
4^1H_g	4.96	250	4777.15	H → L + 2		
4^1G_g	4.99	248	4777.09	H → L + 2		
5^1G_g	5.01	248	4754.47	H − 2 → L + 1		
5^1T_{1g}	5.07	245	4778.31	H → L + 2		
4^1T_{2g}	5.13	242	4778.54	H → L + 2		
3^1T_{1u}	5.22 (10.388)	238	4754.91	H → L + 7	4.84,[g] 4.92[h] (E)	4.48 (3^1T_{1u})
5^1H_g	5.25	236	4754.48	H − 2 → L + 1		
7^1H_u	5.39	230	4771.79	H → L + 9		
5^1T_{2u}	5.44	228	4783.71	H → L + 4		
6^1G_u	5.50	225	4784.57	H → L + 4		
8^1H_u	5.51	225	4784.93	H → L + 4		
7^1G_u	5.55	224	4785.15	H → L + 4		
4^1T_{1u}	5.56 (0.052)	223	4784.97	H → L + 4		5.02 (4^1T_{1u})
9^1H_u	5.61	221	4784.93	H → L + 4		5.10 (5^1T_{1u})
2^1A_u	5.61	221	4785.36	H → L + 4		
5^1T_{2g}	5.61	221	4754.41	H − 3 → L		
6^1G_g	5.71	217	4755.23	H − 2 → L + 9, H − 2 → L + 3		
6^1T_{1g}	5.73	217	4754.82	H − 5 → L		
6^1H_g	5.75	216	4754.84	H − 3 → L		
7^1H_g	5.76	215	4755.21	H − 1 → L + 9, H − 1 → L + 3		
7^1G_g	5.78	215	4754.78	H − 5 → L		
8^1H_g	5.93	209	4789.50	H → L + 5		
6^1T_{2g}	5.94	209	4790.77	H → L + 5		
5^1T_{1u}	5.95 (0.606)	208	4754.44	H − 1 → L + 6	∼5.5[f,g] (F)	5.47 (6^1T_{1u})
7^1T_{1g}	5.96	208	4790.43	H → L + 5		
7^1T_{2g}	6.01	206	4789.70	H → L + 5		
6^1T_{1u}	6.03 (0.848)	206	4754.48	H − 1 → L + 6		
8^1G_u	6.06	204	4754.59	H − 1 → L + 6, H − 6 → L		
10^1H_u	6.26	198	4754.75	H − 6 → L		
6^1T_{2u}	6.33	196	4754.80	H − 6 → L		
					5.90,[g] 5.94[h] (G)	5.72 (7^1T_{1u})
						5.98 (8^1T_{1u})

[a]TDDFT result (Ref. 18).
[b]Oscillator strength is given in parentheses for optically allowed states.
[c]The $\langle r^2 \rangle$ value in the ground state is 4755.45 (Å^2).
[d]HOMO.
[e]LUMO.
[f]Observed absorption spectrum in *n*-hexane (Ref. 17).
[g]Observed absorption spectrum in *n*-pentane (Ref. 29).
[h]Observed absorption spectrum in gas phase 557 °C (Ref. 30).

number of calculated states given in Table II is 61 in I_h symmetry, which corresponds to 230 states in the C_{2h} symmetry used in the actual calculation in which the degeneracy was not considered. Comparisons with experimental findings[17,29,30] and previous TDDFT results[18] are also given in Table II. Figure 1 shows the calculated absorption spectrum of C_{60} compared with the experimental spectrum in *n*-heptane.[29] For optically allowed $^1T_{1u}$ states, the SAC-CI results show good agreement with the experimental spectrum, although the SAC-CI results slightly overestimated the excitation energies in the range 0.1–0.2 eV.

In the visible region (700–430 nm and 1.77–2.9 eV), very weak absorptions termed β, γ, and δ are observed.[16]

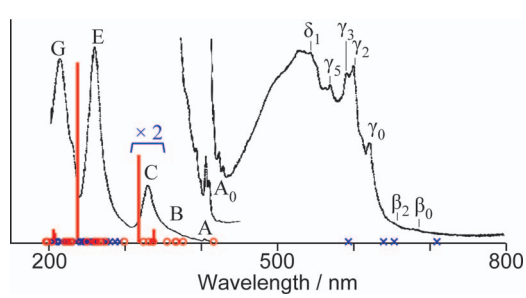

FIG. 1. Experimental absorption spectrum of C_{60} in *n*-heptane (Ref. 29) and the SAC-CI computational results. The vertical lines represent optically allowed states. The cross symbols and open circles denote optically forbidden gerade (even) and ungerade (odd) states, respectively.

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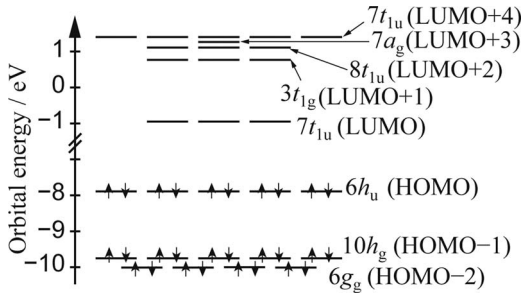

FIG. 2. Valence orbital structure of C_{60} in I_h symmetry.

The present SAC-CI calculation showed that the lowest singlet excited state was 1^1G_g. The peaks in this region have rich vibronic structures, which have been studied using fluorescence measurement techniques.[2] The absorptions in this region would correspond to the HOMO–LUMO transitions. The valence orbital diagram is shown in Fig. 2. The HOMO to LUMO transitions, $6h_u \rightarrow 7t_{1u}$ give 1^1T_{1g}, 1^1T_{2g}, 1^1G_g, and 1^1H_g states.[31] These four states that are degenerated in the orbital energy level are split by configuration interaction. By our CIS calculation, the energy levels of the excited states were 1^1T_{2g} (3.04 eV) < 1^1T_{1g} (3.31 eV) < 1^1G_g (3.47 eV) < 1^1H_g (3.94 eV). These CIS results agreed with the previous semiempirical CNDO/S calculations.[2,10] At the SAC-CI level, the energy level was altered by the electron correlation effect. The 1^1G_g state was stabilized more by the electron correlation than were the 1^1T_{1g} and 1^1T_{2g} states. Highly degenerated G_g or H_g states may involve larger nondynamical correlation effects than those of the T_{1g} and T_{2g} states. The TDDFT with PBE0 functional calculation, whose results were shown in Table III, also provided the same order of the energy levels as the SAC-CI results in this energy region. However, the energy differences between the states obtained by the SAC-CI were larger than those found by PBE0; the energy differences of 1^1G_g, 1^1T_{1g}, and 1^1T_{2g} were less than 0.01 eV by the PBE0 calculation.

TABLE III. TDDFT results for C_{60} with PBE0 functional. The excitation energies (EE), oscillator strengths (Osc), and the corresponding SAC-CI EE are given.

State	EE (eV)	Osc	SAC-CI (eV)
1^1G_g	2.167	…[a]	1.75
1^1T_{2g}	2.169	…	1.90
1^1T_{1g}	2.172	…	1.94
1^1H_g	2.360	…	2.09
1^1T_{2u}	3.314	…	2.98
1^1H_u	3.314	…	3.30
1^1G_u	3.394	…	3.38
2^1G_u	3.563	…	3.39
2^1H_u	3.613	…	3.50
1^1T_{1u}	3.628	0.057	3.67
3^1H_u	3.649	…	3.68
3^1G_u	3.694	…	3.69
2^1T_{2u}	3.721	…	3.74
3^1T_{2u}	3.778	…	3.83
2^1T_{1u}	4.033	0.603	3.90
2^1T_{1g}	4.454	…	4.32

[a]Optically forbidden.

The lowest singlet excited state was suggested to be the 1^1T_{1g} by the MCD spectroscopic study,[14] because a B-term originating in the $^1T_{1g}$ states was observed at 640 nm (1.94 eV). The present study calculated the 1^1T_{1g} state at 1.94 eV; therefore, our result does not contradict the MCD result because other forbidden states are not observed by the MCD spectroscopy. The lowest singlet excited state was assigned to the 1^1T_{1g} state by the analyses of the vibronic structure of fluorescence and fluorescence-excitation spectra.[10–13] Our result is different from the results of these studies, in which experimental spectra were analyzed based on the Herzberg–Teller couplings using the CNDO/S wavefunctions.[10] These studies showed that experimental spectra can be explained based on the vibronic intensities calculated with CNDO/S wavefunctions if one assumed that the 1^1T_{1g} is the lowest state. The CNDO/S calculation, however, failed to reproduce the energy order of the excited states. Sassara *et al.* simulated the fluorescence spectrum using the calculated vibronic transition intensities and the observed spectrum was well reproduced by mixing the 1^1T_{1g}, 1^1T_{2g}, and 1^1G_g states; these three states have very close energy.[13] The SAC-CI calculation can describe the stabilization of the 1^1G_g states; that is necessary to explain the quasi-degeneracy of the 1^1T_{1g}, 1^1T_{2g}, and 1^1G_g states. Although this stabilization of the 1^1G_g state is reasonable, the SAC-CI seems to underestimate the excitation energy of the 1^1G_g state in comparison with the experimental absorption spectrum. It is unclear whether the deviation is attributed to the insufficient computational conditions (basis sets, molecular geometry, perturbation selection, etc.) or the limitation of the SAC-CI method with singles and doubles excitations. To analyze the vibronic structure of the fluorescence spectrum more precisely than did the previous semiempirical calculations, we must calculate the vibronic coupling terms based on the SAC-CI results. Such analyses of the vibronic structure based on the SAC-CI wavefunctions are beyond the purpose of the present study. We could not find other researches for the lowest excited states of C_{60} except for the studies based on the CNDO/S wavefunction. We cannot conclude that the lowest singlet excited state of C_{60} is the 1^1G_g based on the present results only. However, we found that the electron correlation effect must be essentially important in determining the energy levels of these four low-lying states. It is necessary to verify the reliability of the vibronically induced intensities of these states calculated with the semiempirically obtained wavefunctions.

Around 400 nm (3.10 eV), a weak peak, termed band A, is observed. The HOMO to LUMO+1 transitions would appear at about this level; the transitions of $6h_u \rightarrow 3t_{1g}$ give 1^1T_{1u}, 1^1T_{2u}, 1^1G_u, and 1^1H_u states. The R2PI spectroscopic study suggested that this band is a vibronically induced transition of an optically forbidden state.[16] On the other hand, semiempirical[17] and TDDFT[18] calculations assigned this band to the optically allowed 1^1T_{1u} state. By using the SAC-CI method, the 1^1T_{1u} state was calculated at 3.67 eV. If we assigned this state to band A, the excitation energy error reached 0.6 eV. However, such a large deviation cannot be

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explained at the present level of calculation. The calculated high intensity of the 1^1T_{1u} state, which is about one-sixth of the intensity of 2^1T_{1u}, also contradicts the observations. Therefore, the 1^1T_{1u} state should be assigned to band B, which is a shoulder observed around 3.5 eV (350 nm); band A can be assigned to the optically forbidden 1^1T_{2u} state calculated at 2.98 eV. Then, the observed absorption spectrum can be consistently explained by the SAC-CI calculation. By the SAC-CI calculations, the 1^1T_{2u} state is well separated from the next 1^1H_u and 1^1G_u states; therefore, the vibronic transitions of the 1^1T_{2u} state can be observed as an isolated band. The possible vibronic transitions of the 1^1H_u and 1^1G_u states[32] should appear in a higher energy region than the energy of band A. However, such transitions would be covered by the absorption of the 1^1T_{1u} state and would be observed as band B. We obtained similar results with using the PBE0 calculation; however, the PBE0 calculation underestimated the energy difference between 1^1T_{2u} and 1^1H_u in comparison with the SAC-CI results (see Table III). The present SAC-CI results of 400–300 nm showed better agreement with the experimental absorption spectrum than the previous semiempirical[17] and TDDFT[18] calculations in terms of the peak position and intensity, and are consistent with the R2PI spectroscopic study. Significant functional dependence was reported for the excitation energies of C_{60} fullerene.[33] Pure (nonhybrid) TDDFTs underestimated the excitation energy and oscillator strength of the 1^1T_{1u} state. It was coincidentally close to the experimental findings; however, the present study shows that hybrid TDDFT results are more reliable than pure DFTs based on the SAC-CI results.

We assigned the 2^1T_{1u} and 3^1T_{1u} states to bands C and E, respectively. The SAC-CI calculation overestimated the excitation energy of band E about 0.2 eV probably because the present computational conditions were insufficient for these higher excited states. The states that correspond to band G observed at 6.53 eV (190 nm) were not obtained by the present SAC-CI calculation, probably because the number of initial vectors was insufficient (the 80 solutions of the CIS from the lowest energy were used for each symmetry). Based on the calculated electron spatial extents $\langle r^2 \rangle$, the low-lying states that have Rydberg character were the 4^1H_g, 4^1G_g, 5^1T_{1g}, and 4^1T_{2g} states calculated around 5.0 eV (248 nm). The low-lying optically allowed state with Rydberg character was the 4^1T_{1u} state; its intensity was very low.

IV. SUMMARY

In summary, a reliable assignment of the absorption spectrum of C_{60} fullerene was given by an *ab initio* quantum chemical method based on a cluster expansion theory of the SAC and SAC-CI methods. The state-of-the-art computation of large molecules is nowadays practicable because of theoretical developments and parallel computing. Electron correlations described by many-body wavefunctions were important for this system, in which many transitions between degenerated orbitals exist

The electronic excited states of C_{60} below 6.2 eV (200 nm) were calculated including optically forbidden states. The calculated lowest singlet excited state was 1^1G_g using the SAC-CI method. The present result is different from the previous assignment based on the analyses of vibronic bands using the CNDO/S wavefunctions. Although, there is room for improvement in the computational conditions of the present results, we found that the electron correlation effect is important in determining the energy levels of the four low-lying states that have the character of degenerated HOMO–LUMO transition. Therefore, the previous analyses using the CNDO/S wavefunctions might be inadequate. Further studies are necessary for the assignment of the lowest singlet excited state of C_{60}.

The present study showed that the lowest optically allowed state, the 1^1T_{2u}, should be assigned to band B observed around 3.5 eV. Band A, a weak absorption around 3.08 eV, would correspond to the optically forbidden 1^1T_{2u} state. The present SAC-CI results in this region showed better agreement with the experimental absorption spectrum than the previous calculations. The present assignments were consistent with the previous R2PI spectroscopic study. The presented SAC-CI results were qualitatively consistent with hybrid TDDFT results rather than pure TDDFT ones.

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